

Topic-1**Expert System**

It is a computer program or computer system that performs a job that would otherwise be performed by a human expert. For example

1. Medical diagnosis
2. Car engine fault
3. Mineral prospecting
4. Plant prospecting
5. Chess game
6. Oil prospecting
7. Weather forecasting
8. Tax

How expert systems are designed/Analysed/Evaluated/Created?

- Facts and details are collected and recorded about (*plant, disease, car part etc*)
- From user point of view questions and their answers are recorded
- Knowledge base is designed
- Rule base is designed
- Inference Engine is designed
- Interactive (*screens that can communicate with people*) input screens are designed
- Output screens are designed
- Suggestions, solutions, conclusions and treatments from an expert are added
- The new system is tested

Note: above answer is for all types of expert systems

How expert systems will diagnose/work/evaluate/programmed/Explain input, processing and output

Input: Interactive Input screen pops out

Expert system will then ask questions and user will type in the answers
Symptoms and samples are also added

Processing: Expert system will then analyze these answers using its knowledge
-base and rule base with the help of inference engine

Output:

- Disease is diagnosed and treatments are suggested
- Minerals are identified and their percentage is calculated
- Car's fault is identified and solutions are given/part to be replaced is suggested
- Oil is identified its depth and geological state is measured
- Plant is identified and its features are given
- Next move to be made is suggested (chess game)